

Experiment. No: 7 Use Aflogical OSE to Extract Data from an Android Device

Aim

To use AFLogical OSE (Open Source Edition) to perform logical extraction of forensic data from an Android device and recover information such as SMS messages and call logs.

Objective

- To understand Android logical data extraction.
- To configure the Android Debug Bridge (ADB) environment.
- To connect an Android device to the computer.
- To install and use AFLogical OSE.
- To extract SMS messages from an Android device.
- To extract call log information from an Android device.
- To examine the extracted forensic data.

Software and Requirements

- Windows Operating System
- Android smartphone
- AFLogical OSE
- Android Debug Bridge (ADB)
- Android Platform Tools
- Java Development Kit (JDK)
- USB cable
- USB Debugging enabled on the Android device
- Text editor for viewing extracted data

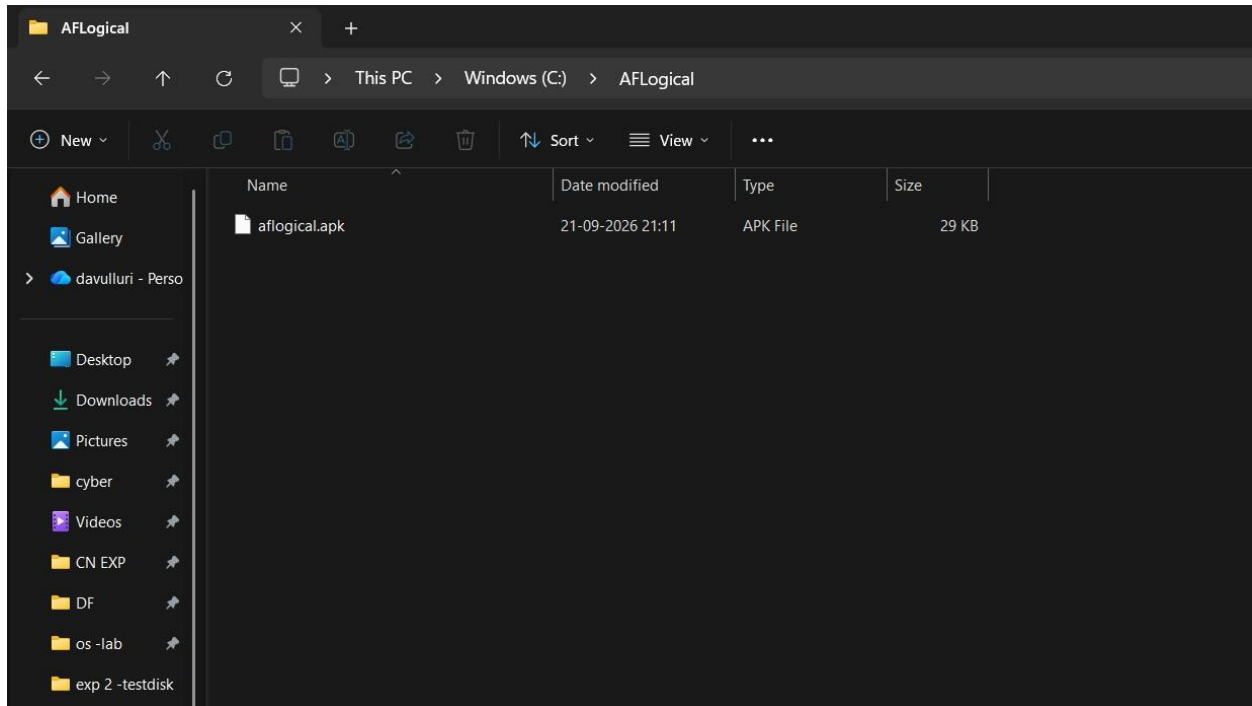
Introduction

AFLogical OSE is an Android forensic data extraction tool used for performing logical acquisition of information from Android devices. It can be used to extract different types of user data, including call logs, SMS messages, contacts, and MMS-related information. The extracted information can then be examined for digital forensic investigation.

Procedure

Step 1 – Prepare the AFLogical OSE Application

The AFLogical OSE application package was obtained and placed in a working folder on the computer. The working folder contained the AFLogical application package required for installation on the Android device.



Step 2 – Verify Java Installation

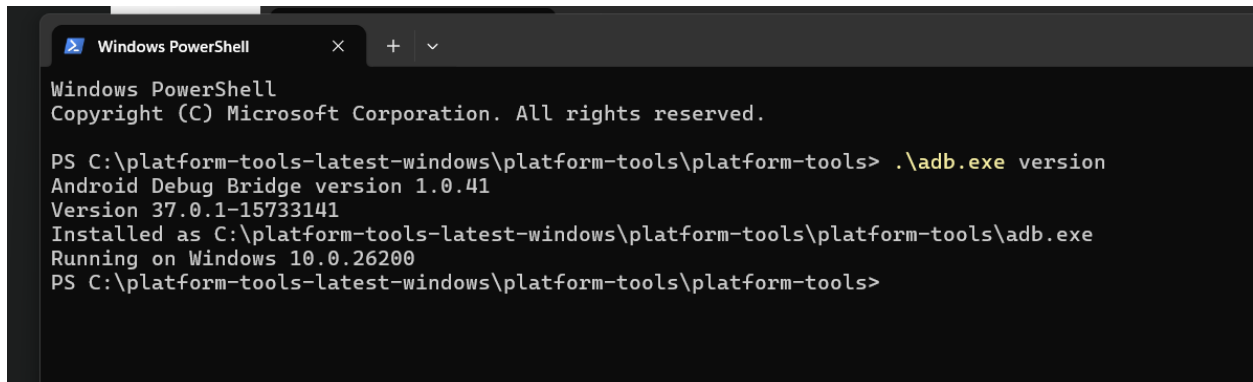
Java was installed and verified on the computer. The installed Java environment was successfully detected and was ready for use with the Android forensic tools.

```
C:\Users\logesh>java -version
java version "26.0.2" 2025-01-17
Java(TM) SE Runtime Environment (build 26.0.2+9-1232)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.2+9-1232, mixed mode, sharing)

C:\Users\logesh>
```

Step 3 – Verify Android Debug Bridge

Android Platform Tools were configured on the computer. ADB was successfully detected and the Android Debug Bridge environment was ready for communication with the Android device.

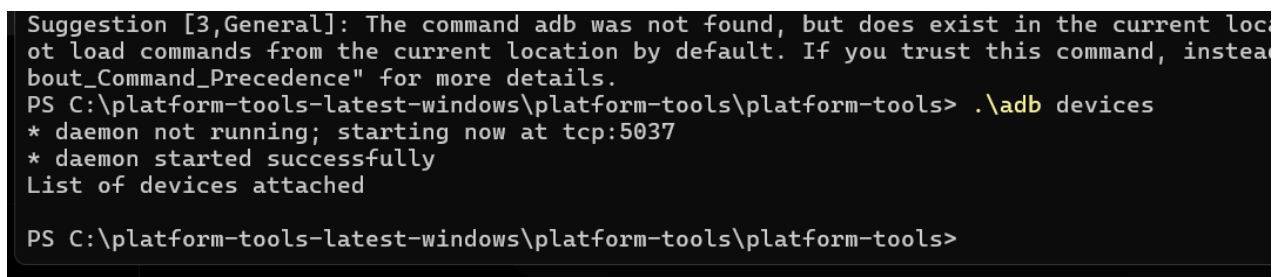


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\platform-tools-latest-windows\platform-tools\platform-tools> .\adb.exe version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\platform-tools-latest-windows\platform-tools\platform-tools\adb.exe
Running on Windows 10.0.26200
PS C:\platform-tools-latest-windows\platform-tools\platform-tools>
```

Step 4 – Connect the Android Device

The Android device was connected to the computer using a USB cable. USB Debugging was enabled on the Android device. The device was successfully detected by ADB.

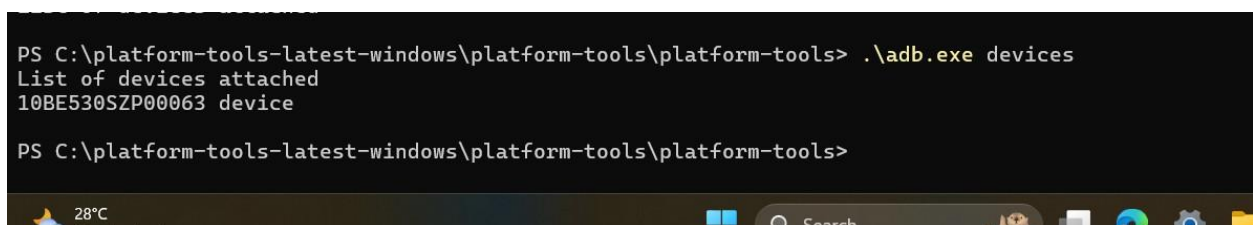


```
Suggestion [3,General]: The command adb was not found, but does exist in the current location. Please use the full path to the command. For more details, see "about_Command_Precedence" for more details.
PS C:\platform-tools-latest-windows\platform-tools\platform-tools> .\adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached

PS C:\platform-tools-latest-windows\platform-tools\platform-tools>
```

Step 5 – Install AFLogical OSE

The AFLogical OSE application was installed on the connected Android device. The installation completed successfully, confirming that the application was successfully transferred and installed on the device.



```
PS C:\platform-tools-latest-windows\platform-tools\platform-tools> .\adb.exe devices
List of devices attached
10BE530SZP00063 device

PS C:\platform-tools-latest-windows\platform-tools\platform-tools>
```

Step 6 – Extract Android Forensic Data

After installation, the AFLogical OSE application was used to obtain logical forensic information from the Android device. The extracted data included SMS messages and call logs. The extracted information was saved as text files on the computer for further examination.

```
C:\Users\logesh\Downloads\platform-tools-latest-windows\platform-tools>adb install com.viaforensics.android.aflogical_ose_1.5.2_OSE-18
Performing Streamed Install
Success
```

Step 7 – Examine Call Log Data

The extracted call log information was opened and examined. The recovered data contained information associated with phone calls, including phone numbers, call duration, call date and time, call type, contact information, and other available call-related metadata.

```
Found 31
PS C:\platform-tools-latest-windows\platform-tools\platform-tools> .\adb.exe shell content query --uri content://sms > sms.txt
PS C:\platform-tools-latest-windows\platform-tools\platform-tools> .\adb.exe shell content query --uri content://call_log/calls > call_logs.txt
PS C:\platform-tools-latest-windows\platform-tools\platform-tools>
```

Step 8 – Examine SMS Data

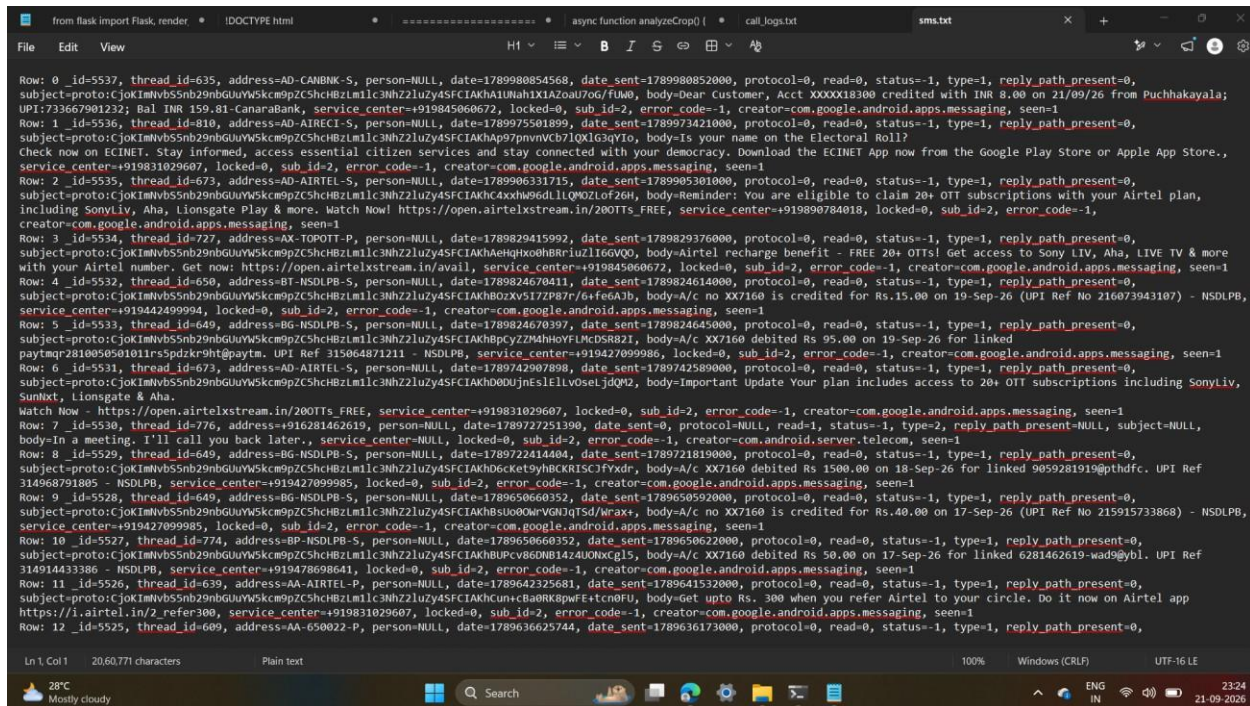
The extracted SMS information was opened and examined. The recovered SMS records contained information such as sender or recipient address, message content, date and time, message identifiers, message status, and service information.

Screenshot 8 – Extracted SMS Data

```
File Edit View H1 ~ H ~ B I S @ ~ 100% Windows (CRLF) UTF-16 LE
Row: 0 formatted_number=NULL, duration=23, subject=NULL, is_call_log_phone_account_migration_pending=0, encrypt=0, is_smart_answer=0, subscription_id=2, photo_id=4913602,
post_dial_digits=, call_screening_app_name=NULL, priority=0, number=916281421823, countryiso=IN, smart_answer_data=NULL, recent=1, smart_answer_record_id=1, photo_uri=NULL,
geoencoded_location=India, vtcalls=1, missed_reason=0, is_business_call=0, block_reason=0,
subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0, add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL,
features=68, transcription=NULL, record_duration=NULL, last_modified=1772165967518, _id=22962, new=1, cnap_logesh, date=1772165928362, name_logesh, type=2, name_logesh, type=1, simid=2,
is_child=0, is_multi=0, call_category=0, carrier=1, smart_record_data=NULL, presentation=1, via_number=, channel=0, number_label=NULL, normalized_number=916281421823,
composer_photo_uri=NULL, phone_account_address=6382435462, phone_account_hidden=0, lookup_uri=NULL, record_path=NULL, voicemail_uri=NULL, matched_number=NULL, smart_translate_data=NULL,
extension_label=NULL, carrier_extension_label=NULL, transcription_state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted_display_name=NULL,
call_screening_component_name=NULL, number_mark_label=254, is_read=NULL
Row: 1 formatted_number=NULL, duration=18, subject=NULL, is_call_log_phone_account_migration_pending=0, encrypt=0, is_smart_answer=0, subscription_id=2, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=916309975365, countryiso=IN, smart_answer_data=NULL, recent=1, smart_answer_record_id=1, photo_uri=NULL, geoencoded_location=India,
vtcalls=1, missed_reason=0, is_business_call=0, block_reason=0, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=68, transcription=NULL, record_duration=NULL, last_modified=1772166034112, _id=22963, new=1, cnap_logesh,
date=1772165975388, name_logesh, type=2, simid=2, is_child=0, is_multi=0, call_category=0, carrier=1, smart_record_data=NULL, presentation=1, via_number=, channel=0, number_label=NULL,
normalized_number=916309975365, composer_photo_uri=NULL, phone_account_address=6382435462, phone_account_hidden=0, lookup_uri=NULL, record_path=NULL, voicemail_uri=NULL,
matched_number=NULL, smart_translate_data=NULL, extension_label=NULL, carrier_extension_label=NULL, transcription_state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL,
asserted_display_name=NULL, call_screening_component_name=NULL, number_mark_label=254, is_read=NULL
Row: 2 formatted_number=NULL, duration=18, subject=NULL, is_call_log_phone_account_migration_pending=0, encrypt=0, is_smart_answer=0, subscription_id=2, photo_id=4913602,
post_dial_digits=, call_screening_app_name=NULL, priority=0, number=91910284721823, countryiso=IN, smart_answer_data=NULL, recent=1, smart_answer_record_id=1, photo_uri=NULL, geoencoded_location=India,
vtcalls=1, missed_reason=0, is_business_call=0, block_reason=0, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=68, transcription=NULL, record_duration=NULL, last_modified=1772166040130, name_logesh, type=2, simid=2, is_child=0,
is_multi=0, call_category=0, carrier=1, smart_record_data=NULL, presentation=1, via_number=, channel=0, number_label=NULL, normalized_number=91910284721823, composer_photo_uri=NULL,
phone_account_address=6382435462, phone_account_hidden=0, lookup_uri=NULL, record_path=NULL, voicemail_uri=NULL, matched_number=NULL, smart_translate_data=NULL, extension_label=NULL,
carrier_extension_label=NULL, transcription_state=0, data_usage=NULL, ai_anti_fraud_data=NULL, location=NULL, asserted_display_name=NULL, call_screening_component_name=NULL,
number_mark_label=254, is_read=NULL
Row: 3 formatted_number=NULL, duration=44, subject=NULL, is_call_log_phone_account_migration_pending=0, encrypt=0, is_smart_answer=0, subscription_id=2, photo_id=0, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=9190808070953, countryiso=IN, smart_answer_data=NULL, recent=1, smart_answer_record_id=1, photo_uri=NULL, geoencoded_location=India,
vtcalls=1, missed_reason=0, is_business_call=0, block_reason=0, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=68, transcription=NULL, record_duration=NULL, last_modified=1772166207895, _id=22965, new=1, cnap_logesh,
date=1772166178229, name_logesh, type=1, simid=2, is_child=0, is_multi=0, call_category=0, carrier=1, smart_record_data=NULL, presentation=1, via_number=, channel=0,
number_label=NULL, normalized_number=9190808070953, composer_photo_uri=NULL, phone_account_address=6382435462, phone_account_hidden=0, lookup_uri=NULL, record_path=NULL,
voicemail_uri=NULL, matched_number=NULL, smart_translate_data=NULL, extension_label=NULL, carrier_extension_label=NULL, transcription_state=0, data_usage=NULL, ai_anti_fraud_data=NULL,
location=NULL, asserted_display_name=NULL, call_screening_component_name=NULL, number_mark_label=254, is_read=NULL
Row: 4 formatted_number=NULL, duration=6, subject=NULL, is_call_log_phone_account_migration_pending=0, encrypt=0, is_smart_answer=0, subscription_id=2, photo_id=4913602, post_dial_digits=,
call_screening_app_name=NULL, priority=0, number=916281421823, countryiso=IN, smart_answer_data=NULL, recent=1, smart_answer_record_id=1, photo_uri=NULL, geoencoded_location=India,
vtcalls=1, missed_reason=0, is_business_call=0, block_reason=0, subscription_component_name=com.android.phone/com.android.services.telephony.TelephonyConnectionService, is_secret=0,
add_for_all_users=0, numbertype=2, conf_callid=NULL, attribution=NULL, features=68, transcription=NULL, record_duration=NULL, last_modified=1772166261613, _id=22966, new=1, cnap_logesh,
date=1772166238851, name_logesh, type=2, simid=2, is_child=0, is_multi=0, call_category=0, carrier=1, smart_record_data=NULL, presentation=1, via_number=, channel=0,
number_label=NULL, normalized_number=916281421823, composer_photo_uri=NULL, phone_account_address=6382435462, phone_account_hidden=0, lookup_uri=NULL, record_path=NULL,
```

Step 9 – Detailed Examination of Extracted Data

The extracted forensic records were further examined to understand the available information. The SMS records contained detailed metadata and message contents. This demonstrated that logical acquisition can recover useful information from an Android device for forensic examination.



```
Row: 0 id=5537, thread_id=635, address=AD-CANBANK-S, person=NULL, date=1789980854568, date_sent=1789980852000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHAIUNahIXIAZoaU7oG/fWu0, body=Dear Customer, Acct XXXXX18300 credited with INR 8.00 on 21/09/26 from Puchhakaya; UPI:733667901232; Bal INR 159.81-CanaraBank, service_center=+919845060672, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 1 id=5536, thread_id=810, address=AD-AIRECI-S, person=NULL, date=1789975501899, date_sent=1789973421000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHAp97pnnVCb7lQkIC3qVt0, body=Is your name on the Electoral Roll? check now on ECINET. Stay informed, access essential citizen services and stay connected with your democracy. Download the ECINET App now from the Google Play Store or Apple App Store., service_center=+919831029607, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 2 id=5535, thread_id=673, address=AD-AIRTEL-S, person=NULL, date=1789906331715, date_sent=1789905301000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHCAxxhW96dLLQWZlOf26H, body=Reminder: You are eligible to claim 20+ OTT subscriptions with your Airtel plan, including SonyLiv, Aha, Lionsgate Play & more. Watch Now! https://open.airtelxstream.in/200TTs.FREE, service_center=+919890784018, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 3 id=5534, thread_id=727, address=AX-TOPTT-P, person=NULL, date=1789829415992, date_sent=1789829376000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHAEHqX00hBRrIuZl16GVQ0, body=Airtel recharge benefit - FREE 20+ OTTs! Get access to Sony LIV, Aha, LIVE TV & more with your Airtel number. Get now: https://open.airtelxstream.in/avail, service_center=+919845060672, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 4 id=5532, thread_id=658, address=BT-NSDLPB-S, person=NULL, date=1789824670411, date_sent=1789824614000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKH80ZxY517Z987r/Gf6a3b, body=A/c no XX7160 is credited for Rs.15.00 on 19-Sep-26 (UPI Ref No 216073943107) - NSDLPB, service_center=+919442499994, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 5 id=5533, thread_id=649, address=BG-NSDLPB-S, person=NULL, date=1789824670397, date_sent=1789824645000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHBPcyZ7M4hhoVfLMCD5R821, body=A/c XX7160 debited Rs 95.00 on 19-Sep-26 for linked paytmr2810050501011rs5pdzkr9ht@paytm. UPI Ref 315064871211 - NSDLPB, service_center=+919427099986, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 6 id=5531, thread_id=673, address=AD-AIRTEL-S, person=NULL, date=1789742907898, date_sent=1789742589000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKH80DUjNeIsEl1V0seIjdnQ2, body=Important Update Your plan includes access to 20+ OTT subscriptions including SonyLiv, SunNxt, Lionsgate & Aha. Watch Now - https://open.airtelxstream.in/200TTs.FREE, service_center=+919831029607, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 7 id=5530, thread_id=776, address=+916281462619, person=NULL, date=17897272261390, date_sent=0, protocol=NULL, read=1, status=-1, type=2, reply_path_present=NULL, subject=NULL, body=In a meeting. I'll call you back later., service_center=NULL, locked=0, sub_id=2, error_code=-1, creator=com.google.android.server.telecom, seen=1
Row: 8 id=5529, thread_id=649, address=BG-NSDLPB-S, person=NULL, date=1789722414404, date_sent=1789721819000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKH8Gckct9yhBCKRISCFYxdr, body=A/c XX7160 debited Rs 1500.00 on 18-Sep-26 for linked 9059281919@pthdfc. UPI Ref 314968791805 - NSDLPB, service_center=+919427099986, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 9 id=5528, thread_id=649, address=BG-NSDLPB-S, person=NULL, date=1789650660352, date_sent=1789650592000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKH8U0u0MVGnJqTsd/Wraxe, body=A/c no XX7160 is credited for Rs.40.00 on 17-Sep-26 (UPI Ref No 215915733868) - NSDLPB, service_center=+919427099986, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 10 id=5527, thread_id=774, address=BP-NSDLPB-S, person=NULL, date=1789650660352, date_sent=1789650622000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKH8UPCV86DNB14Z4UONXGp15, body=A/c XX7160 debited Rs 50.00 on 17-Sep-26 for linked 6281462619-wad9@ybl. UPI Ref 31491443386 - NSDLPB, service_center=+919478690641, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 11 id=5526, thread_id=630, address=AA-AIRTEL-P, person=NULL, date=1789642325681, date_sent=1789641532000, protocol=0, read=0, status=-1, type=1, reply_path_present=0, subject-proto:CjokImvB55nb29nbgUuV5Kcm9pZC5hCHBzLmlc3NhZ2luZy45FCIAKHcunCBA8RK8pWfEtCn0F0, body=Get upto Rs. 300 when you refer Airtel to your circle. Do it now on Airtel app https://i.airtel.in/2_refer300, service_center=+919831029607, locked=0, sub_id=2, error_code=-1, creator=com.google.android.apps.messaging, seen=1
Row: 12 id=5525, thread_id=609, address=AA-650022-P, person=NULL, date=1789636625744, date_sent=1789636173000, protocol=0, read=0, status=-1, type=1, reply_path_present=0,
```

Observations

1. Java was successfully installed and detected.
2. Android Platform Tools were successfully configured.
3. ADB successfully detected the connected Android device.
4. AFLogical OSE was successfully installed on the device.
5. Logical forensic data was successfully extracted.
6. Call log information was recovered.
7. SMS information was recovered.
8. The extracted information contained useful metadata for forensic analysis.

Result

AFLogical OSE was successfully used to perform logical extraction from an Android device. The experiment successfully recovered call logs and SMS messages, which were examined as part of the forensic analysis.

Conclusion

This experiment demonstrated the use of AFLogical OSE for Android logical forensic extraction. The Android device was successfully connected to the computer, AFLogical OSE was installed, and forensic information including SMS messages and call logs was successfully extracted and examined. The experiment provided practical knowledge of Android device acquisition and the examination of extracted mobile forensic data.